

WAVE ENERGY DEVICE

Background

With an increased demand for producing energy efficiently and environmentally friendly, wave power may, with improved technology, represent a larger proportion of produced electric energy. ~~There are today turbines~~ today's turbines are placed in the sea; ~~but, Still,~~ these are, in many cases, exposed to hard conditions due to wind and hard waves, which may damage the turbines. There are devices in which turbines are protected, e.g., overtopping wave energy converters. In such devices, water from waves is collected and enters an elevated tank ~~which uses the static pressure of the elevated water to rotate a turbine located at the bottom of it and thereafter,~~ which uses the static pressure of the elevated water to rotate a turbine located at the bottom of it; thereafter, the water returns to the sea through an outlet. One deficit with this type of devices is that only the gravitational potential energy of the wave is utilized. Moreover, the wave needs to be of a specific height to enter the elevated tank. There is, therefore, a need for a device in which wave energy is utilized with improved efficiency and for which turbines also are protected.

Summary

It is an object of the present invention to provide a solution ~~which that~~ mitigates the above-mentioned problems. Further, it is an object to provide a wave energy device for generating electrical energy from wave power and a method for concentrating the energy of a wave using said device. It is also an object to provide a method of concentrating the energy of a wave and generating electric power from the concentrated wave.

According to one aspect of the present invention, there is provided a wave energy device for generating electrical energy from wave power. The device comprises at least one wave power turbine arranged in the sea to generate electricity from the motion of water in the sea. It further comprises two side walls arranged on two sides of the wave

power turbine, defining a main water channel allowing water to flow between the two side walls in the main water channel and two guiding walls, each connected to a side wall at ~~a-the~~ first end. The distance between the guiding walls is increasing in the direction towards a second end being opposite said first end. ~~The-a~~ At least one wave power turbine is oriented in the main water channel to receive water flowing substantially in a wave propagation direction and with concentrated flow from the two guiding walls.

With the disclosed device, the kinetic energy of the water in a wave may be concentrated and utilized. The space between the guiding walls at the second end may form a wave inlet of the main water channel. The space between the side walls downstream in the main wave direction, away from the inlet, may form an outlet of the main water channel. A large amount of water in the wave may enter the inlet of the wave energy device between the guiding walls. Thereby, the wave energy may be collected and concentrated, by continuously reducing the cross-sectional area of the main water channel, through which the wave is guided, until reaching the space between the side walls. Based on the principle of continuity, the wave speed will increase if the amount of water is kept constant.

For instance, for a wave, about 25 % of the water entering the main water channel through its inlet between the guiding walls may exit the wave energy device through the outlet between the side walls, in one embodiment of the invention. The water may leave the wave energy device through an outlet with a smaller ~~cross-cross-~~section than that of the inlet.

In some embodiments, the proportion between the outlet surface area and the inlet surface area is smaller than the proportion of the incoming water amount and the amount of water that leaves the device through the outlet.

The energy of a wave that is concentrated to a smaller ~~cross-cross-~~section may, to the extent allowed by the number and type of turbines of the wave energy device, be utilized for the generation of electric energy. The disclosed wave energy device may utilize wave energy more effectively than prior art wave energy devices.

The side walls and the guiding walls may be arranged with vertical surfaces, allowing waves, which mainly propagate horizontally, to move between the walls. Alternatively, the side walls and/or guide walls may be inclined inwards at a top portion towards ~~the~~ at least one turbine so ~~as that~~ there is a difference between the distance
 5 between the bottom of two opposing walls and the distance between the top of two opposing walls.

In one embodiment, the side walls may be portions of a common structure. For instance, the side walls may together form a half-circular or half-elliptic pipe in which the turbine may be located.

10 The disclosed wave energy device may utilize both the kinetic and potential energy of the waves. The present invention is not dependent on building an elevated tank for storing water. The present invention may further be able to utilize waves of practically all heights. The disclosed wave energy device may withstand and utilize waves with high energy, even such as those caused by hurricanes, without the wave energy
 15 device being damaged. The wave energy device may have been found to work successfully through experiments.

In one embodiment, the guiding walls are supported by support elements, such as concrete, stones, or other structures, on the opposite side relative to the side that is exposed to the main force from the waves, so as to protect the walls from collapsing.

20 Such an embodiment may further increase the capacity ~~for of~~ the device to withstand large forces.

In one embodiment ~~the~~, at least one turbine may be placed adjacent to the space between the guiding walls, so that the wave, after passing the guiding walls in a downstream direction, immediately reaches ~~the~~ at least one turbine.

25 In one embodiment, the side walls extend in parallel, or at least substantially parallel.

Parallel side walls may be suitable for placing turbines between said walls. The side walls being parallel may also be suitable to not expose the walls ~~for to~~ unnecessary high forces since they may form a portion of the wave energy device where the wave
 30 propagation speed is as highest. When the wave has passed ~~the~~ at least one turbine,

there may be of no importance to further concentrate the wave, and thereby it may be without benefit, to also have narrowing side walls.

In one embodiment, the side walls and the guiding walls are made of concrete, plastic or metal.

Concrete may be a suitable material to withstand large forces from waves that are concentrated by the guiding walls and to be resilient against deterioration due to the sea environment. The strength and the resilience of the wave energy device may be improved by selecting some concrete with sufficient strength, which may depend on the raw materials used and the manufacturing process of the concrete. In one embodiment, the wave energy device further comprises a vault extending at least partly between the two side walls and/or at least partly between the two guiding walls.

A vault is to be understood as an arched form, above the side walls and/or guiding walls which that enclose at least a portion of the main water channel. The side walls may, in such an embodiment, be different portions of one common structure.

The inclusion of a vault in the wave energy device extending between the two side walls and/or the two guiding walls, may imply that water and air inside the wave energy device ~~is-are~~ trapped, and the air is compressed, so that incoming waves move in a piston manner with higher velocity towards and through the turbine. Thereby, the compressed air may reach a velocity of 50 -150 m/s (often about 90 m/s) and the water a velocity of 5-15 m/s (often about 10 m/s) at the outlet of the wave energy device.

With the vault, the turbulent kinetic energy may reach about 150-500 % of the input wave per cross meter, values which have been measured through experiments to 155-475%.

In one embodiment, the vault extends along less than 2/3 of the extension of the guiding walls and side walls, preferably along less than half of the extension of the guiding walls and side walls.

By only letting the vault extend along a portion of the guiding walls and/or the walls, the formation of large air bubbles under the vault, which can have a negative impact on the work of the turbine, may be avoided. When the bubbles are formed, they may escape through a portion along the main water channel of which the vault does not

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extend. The bubbles may thereby avoid becoming very large. With the geometry of this embodiment, the turbulent kinetic energy of the output wave per cross meter has been calculated to be between 200-517% of the TKE of the input wave per cross meter (averaged value= **about** 288%). Said geometry also reduces the forces acting on the wave energy device compared to an embodiment in which the vault extends along the entire extent of **the** main water channel.

In one embodiment, the guiding walls are vertically at least as high as the highest point of the vault.

The guiding walls being at least as high as the highest point of the vault may entail that a large fraction of the water which passes under the vault to first move between the guiding walls. Thereby, the guiding walls may take up a large portion of the force from the waves which the wave energy device is exposed to, protecting the vault from very large forces.

In one embodiment, the guiding walls are placed along an inclining bottom surface with a positive inclination in the main water flow direction. The inclination of the bottom surface may cause the incoming waves to rise and the speed of the wave to increase, entailing **a** possible higher effect of the wave energy device.

In one embodiment, the bottom surface of the main water channel may be a seabed.

In one embodiment, the inclination of the leaning bottom surface where the device is arranged is at least 5 degrees, preferably more than 10 degrees, and less than 35 degrees, preferably less than 30 degrees.

In one embodiment, the wave energy device comprises a plurality of turbine generators distributed along a vertical axis, extending between the side walls.

The wave energy device comprising a plurality of wave power turbines may increase the utilization of the incoming wave energy. Especially for high waves, a plurality of wave power turbines may be needed to utilize as much of the wave energy as possible. In the present invention, the number of wave power turbines that could be comprised may depend on the height of the side walls. In one embodiment, the device comprises a 2 m² outlet surface with a single turbine. In other embodiments, the outlet

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surface ~~may be~~ can reach up to 1 m² with a single turbine. It is also possible to have an outlet surface of 2-10 m² and ~~having~~ 2-10 turbines ~~the that jointly are~~ are jointly driven by the water flow through the outlet surface.

In one embodiment, as many wave power turbines as there is space for are distributed along said vertical axis. Hereby, the wave power turbines may be stacked on top of each other.

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In one embodiment, the distance between the side walls is between 0.5 m and 2 m. ~~To have~~ Having a small distance between the walls may allow the incoming wave to be more concentrated compared to a larger distance since the concentration of the waves may depend on the ratio between the ~~cross-cross~~-section at the inlet and the outlet of the main water channel. However, the distance between the walls should be sufficiently large to accommodate ~~the~~ at least one wave power turbine.

In one embodiment, the length of the side walls is between 0.5 m and 2 m.

The length of the walls may be sufficiently large for ~~the~~ at least one wave power turbine to be accommodated between the walls.

In one embodiment, the angle between the guiding walls and the side walls is between 5 and 20 degrees.

A larger angle of the guiding walls may increase the ratio between the ~~cross-cross~~-section of the inlet and outlet of the wave energy device and, thereby, the concentration of the wave energy. On the other hand, an increased angle of the guiding walls, relative to the side walls, will increase forces acting on the wave energy device, implying a trade-off regarding said angle.

In one embodiment, the length of the guiding walls is between 15 m and 300 m.

Longer guiding walls will entail a larger ~~cross-cross~~-section ratio between the inlet and the outlet of the wave energy device and a larger concentration of the wave energy. Longer guiding walls will, however, also entail higher production costs for the wave energy device and a larger force ~~on~~ the wave energy device.

According to a second aspect of the ~~present invention~~ proposed solution, there is provided a use of said wave energy device wherein the device is arranged in a transitional water region.

A transitional water region is to be understood as a region where the depth of the water is between 5 % and 50 % of the wavelength of the present waves. This region has₁ in experiments₁ shown to be the most beneficial ~~to place~~for placing water turbines to utilize wave energy.

5 Further, a concrete base may be beneficial if the wave energy device is placed further out in the sea~~s~~. Such a location may ~~entail~~induce very large forces onto the wave energy device, in particular₁ said concrete base₁ due to a sudden clash of the wave with the concrete base. Moreover, it has been found that such a clash may lead to wave attenuation₁ decreasing the efficiency of the wave energy device.

10 According to a third aspect of the present invention₁ there is provided a method of concentrating the energy of a wave and generating electricity from the concentrated wave using a wave energy device. The method comprises the steps of placing the wave energy device in a sea along a main wave direction, guiding the wave through two guiding walls being tapered in a main water flow direction, increasing the speed of the
15 wave towards a wave power turbine in a main water channel with the tapered guiding walls and generating electricity from the motion of the wave, including kinetic energy, by the at least one wave power turbine.

By the guiding walls being tapered₁ it is meant that the space between the walls is narrowing along a direction so ~~as that~~ the distance between a point on one guiding
20 wall and the closest portion on the other of the guiding walls is narrowing. The walls do not necessarily have to be straight₁ narrowing walls but can be bent or stepwise tapered towards the outlet.

By ~~the this~~ method, the energy of waves at the location of the wave energy device₇ may be efficiently utilized. A wave may enter the main water channel at an inlet
25 between the guiding walls where the space between the guiding walls is as widest. The speed of the wave may increase as the space between the guiding walls ~~is narrowing~~narrows in the wave propagation direction, and thereby the wave energy is concentrated to a smaller ~~cross-cross~~ section area. When the wave has passed the space between the guiding walls₁ it may enter the at least one wave power generator between
30 the walls. Each generator may operate with higher efficiency than if it was placed at the

inlet since the wave energy is concentrated. After the wave has passed the generator, it may leave the main water channel at an outlet between the side walls.

In one embodiment, the height of the wave may increase by contact between the wave and the bottom surface of the wave energy device.

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Brief Description of the Drawings

The invention will in the following be described in more detail with reference to the enclosed drawings, wherein:

Fig. 1 illustrates a front view of a wave energy device according to one embodiment of the invention.

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Fig. 2 illustrates a side view of a wave energy device according to one embodiment of the invention.

Fig. 3a illustrates a top view of a wave energy device according to one embodiment of the invention.

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Fig. 3b illustrates the area of the incoming and outgoing wave.

Fig. 4 illustrates a front view of a wave energy device with a vault according to one embodiment of the invention.

Fig. 5 illustrates a top view of a wave energy device with a vault according to one embodiment of the invention.

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Fig. 6 shows a flow-chart of the method according to one aspect of the invention.

Detailed description

The present invention will be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. In the drawings, like numbers refer to like elements.

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Figs. 1, 2, and 3a illustrates a wave energy device 1. The wave energy device has at least one wave energy turbine 2, which generates electric energy by being rotated by the movement of water, ~~and in~~ particularly waves. Waves may move through the wave energy device with trajectories marked with dashed lines. The average movement defines a main wave direction W. The bottom of the device may be an inclined bottom surface 7, which has an inclination I , which is illustrated in Fig. 2. The inclination I is about 5-15 degrees in some embodiments, preferably about 10 degrees. In some embodiments, the inclined water surface 7 is a concrete base placed onto the seabed. In other embodiments, the inclined bottom surface is a seabed.

In Fig. 3a, the wave energy device is illustrated from above. Parallel with, and on each side of, ~~the~~ at least one turbine on two opposite sides, there are two side walls 3 and 4. The side walls have a length of $L1$ and are spaced apart with a distance of $D1$. In some embodiments, $L1$ is about 0,5-2m or around 1 m. In some embodiments, $D1$ is about 0,5-2m or around 1 m. In other embodiments, the two side walls may not be entirely parallel, but there may be an angle between the side walls.

Connected to each side wall 3 and 4, there is a corresponding guiding wall 5, 6. There is an angle α between the guiding walls and the side wall to which each of the guiding walls is connected, respectively. In some embodiments, the angle α between the side wall and guiding wall is about 15-45 degrees. In the illustrated example, the angle α is about 27.5 degrees. In other embodiments, the angle α may be 20 degrees or even 45 degrees. The angle α is measured as the inner angle between the general extension of the side wall and the extension of the guiding wall, as illustrated in figure 3. That is, how much the guiding walls are "opening up" towards the inlet relative to the extension of the side walls in the tapering form.

The side walls and guiding walls together form a main water channel in which water may flow. The guiding walls have a length $L2$. The length $L2$ of the guiding walls may vary a lot depending on the size of the wave energy device 1. The length may, e.g. for example, be range between 5 and 300 meters. In the illustrated examples, the length of the guiding walls is kept fairly short in order to be able to illustrate the relevant parts in a clear way. In figures, the length $L2$ of the guiding walls is about 6-10 m. In other

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examples the length L2 of the guiding walls may be about 150 meter or even 300 meter long.

Regardless of the length, the main function is the same. ~~the~~ guiding walls 5, 6 form an inlet for water at the opposite side relative the at least one turbine 2. Incoming waves moves through the main water channel and are concentrated by the narrowing of the guiding walls of the main water channel. Due to the inclining bottom surface 7, the elevation of throughgoing waves may increase in the portion where the bottom is inclining. The wave may, after propagating between the guiding walls 5 ~~and~~ 6, reach ~~the at least at least~~ one turbine 2 between the side walls 3 ~~and~~ 4 with higher velocity than when entering the main water channel, and thereby the guiding walls cause more electric energy to be generated. After reaching ~~the~~ at least one turbine 2, the water in the wave may continue out of the device and leave the main water channel through the outlet between the side walls 3, 4.

It is not necessarily all water in the device that enters through the main water channel inlet and leaves it through the main water channel outlet. Some water may enter the main water channel from above. Also, some water may leave the main water channel over the guide walls or side walls since the top of the channel may be open, as it is in the embodiments illustrated in Fig. 1, 2, and 3.

In Fig 4 and Fig. 5, another, entirely combinable embodiment, of the wave energy device is illustrated from a front and a top perspective, respectively. In this illustrated embodiment, there is a vault 8 extending across the main water channel. The vault is connected to the guiding walls 5, 6, and/or the side walls 2 ~~and~~ 3, and extends along a portion of the extension of the main water channel. From the top perspective illustrated in Fig. 5, ~~the~~ at least one turbine 2 is hidden under the vault 8.

In other embodiments, the vault does not cover the turbine but extends along another portion of the main water channel.

The waves which go through the main water channel are confined by the vault. Water can neither leave nor enter the device at a portion of the main water channel, which is covered by the vault 8. Thereby, the vault may contribute to concentrating the

water in throughgoing waves. In some embodiments, a vault extends along the entire portion of the main water channel.

In embodiments with a covering vault 8, all water enters the device at its inlet and leaves through its outlet. The waves are there concentrated by the narrowing of the cross-sectional area, which depends on the height from the inclining bottom surface 7 and the distance between the guiding walls 5, 6. In Fig. 3b, the area of the incoming and outgoing wave is illustrated. The area A_o of the outgoing wave is smaller than the area A_i of the incoming wave, and thereby the velocity ~~mist-must~~ be larger at the outlet of the main water channel due to the continuity of mass.

Fig. 6 illustrates a flow-chart, related to a method of concentrating the energy of a wave and generating electricity from the concentrated wave using an energy device, of which some embodiments are illustrated in Fig. 1-Fig. 5. In the flow-chart, the method steps S1-S4, of said method are represented.

A first step S1 is to place a wave energy device 1, in a sea oriented to receive waves in a main wave direction W. By placing the device along a main wave direction, utilization of energy generation from the waves may be enabled.

The second step S2 is to guide the wave through two guiding walls 5, 6 being tapered in the main wave direction. By tapered, it is meant that the guiding walls are narrowing in one direction, as is best seen in the illustrations in Fig. 3a and Fig. 5.

Incoming waves of the device are confined by the guiding walls to extend with smaller width due to the guiding walls being narrower in the main wave direction.

A further step S3 is to increase the speed of the waves towards a wave power turbine 2 in a main water channel with the tapered guiding walls 5, 6. The speed is increased due to the continuity principle. When waves are confined to extend with smaller width, the speed of the waves increases as the flow is constant.

A step S4 is to generate electricity from the motion of the wave, including the wave's kinetic energy, by ~~the~~ at least one wave power turbine. The energy is, by the previously described step, concentrated ~~to-in~~ a smaller area, and within this area, a turbine is placed, which is moved by the movement of the wave.

All features as disclosed in relation to any of the embodiments of the device are equally applicable ~~on~~to the method as described, and vice versa. E.g. the vault 8, which is used for confining the water in the waves, ~~are~~is equally applicable on a method step of confining the water using a vault 8.

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Claims

1. A wave energy device (1) for generating electrical energy from wave power, said device comprising:

5 at least one wave power turbine (2) arranged in a sea to generate electricity from ~~the~~ motion of water in the sea;

two side walls (3, 4) arranged on two sides of ~~the~~ at least one wave power turbine defining a main water channel allowing water to flow between the two side walls in the main water channel; and

10 two guiding walls (5, 6), each connected to a respective of said side walls (3, 4) at ~~a-the~~ first end and wherein the distance between the guiding walls is increasing in the direction towards a second end being opposite said first end,

and wherein ~~the~~ at least one wave power turbine (2) is oriented in the main water channel to receive water flowing substantially in a main wave propagation direction (W)

15 and with concentrated flow from the two guiding walls (5, 6).

2. A wave energy device (1), according to any preceding claim, further comprising a vault (8) extending at least partly between the two side walls and/or at least partly between the two guiding walls (5, 6).

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3. A wave energy device (1) according to claim 4 claim, wherein the vault extends along less than 2/3 of the extension of the guiding walls (5, 6) and side walls (3, 4), preferably along less than half of the extension of the guiding walls (5, 6) and side walls.

25 4. A wave energy device (1) according to claim 5, wherein the guiding walls (5, 6) vertically extend upwards from a surface level at least 0.5-1.5 times the distance from the surface to the bottom.

5. A wave energy device (1) according to any of the preceding claims, wherein the guiding walls (5, 6) are placed along an inclining bottom surface (7) with a positive inclination in the main water flow direction.

5 6. A wave energy device (1) according to claim 7, wherein the inclination (I) of the inclining bottom surface (7) where the device is arranged is at least 5 degrees, preferably more than 10 degrees and most preferably more than 15 degrees and less than 35 degrees, preferably less than 30 degrees.

10 7. A wave energy device (1) according to any preceding claims, comprising a plurality of turbine generators distributed along a vertical axis, extending between the walls.

15 8. A wave energy device (1) according to any preceding claims, wherein the distance (D1) between the walls is between 0.5 m and 5 m.

9. A wave energy device (1) according to any preceding claims, wherein the length of the side walls (L1) is between 0.5 m and 5 m.

20 10. A wave energy device (1) according to any preceding claims, wherein the angle between the guiding walls (5, 6) and the side walls (3, 4) is between 20 and 40 degrees, and preferably between 25 and 30 degrees.

25 11. A wave energy device (1) according to any preceding claim, wherein the length of the guiding walls (5, 6) is between 15 m and 300m, preferably between 50 and 150 m.

12. Use of a wave energy device (1) according to claim 1-14 wherein the device is arranged- in a transitional water region.

13. A method of concentrating the energy of a wave and generating electricity from the concentrated wave using a wave energy device (1), comprising the steps of:

placing (S1) the wave energy device (1) in a sea along a main wave propagation direction (W) of the wave,

5 guiding (S2) the wave through two guiding walls (5, 6) being tapered in the main wave direction,

increasing (S3) the speed of the wave towards a wave power turbine in a main water channel with the tapered guiding walls (5, 6),

generating (S4) electricity from the motion of the wave, including kinetic energy,

10 by ~~the~~ at least one wave power turbine (2).

14. The method of concentrating the energy of a wave and generating electricity according to to claim 17, further ~~comprising-comprises~~ the step of:

15 guiding the wave between the two guiding walls (5, 6) and a vault at least partially extending between the two guiding walls (5, 6).

Abstract

A wave energy device for generating electrical energy from wave power is presented. The said device comprises at least one wave power turbine (2) arranged in the sea to generate electricity from the motion of water in the sea, two side walls (3, 4) arranged on two sides of the at least one wave power turbine defining a main water channel allowing water to flow between the two side walls in the main water channel and two guiding walls (5, 6) each connected to a side wall (3, 4) at a first end and wherein the distance between the guiding walls is increasing in the direction towards a second end being opposite said first end.

Fig. 5 for abstract.